



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/LMIswitc/366136

Date:

04/08/2023(16:50:46)

Subject:Revision of LMI/OGN/OPS / SYST / 012, SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED

Revision of LMI/OGN/OPS / SYST / 012

SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

OGN/OPS/SYST/012 has been revised by COS in april 2022.No changes envisaged by COS.

Only reviewing done after two years.

Put up for kind approval.

Attachments:

1. LMI/OGN/OPS / SYST / 012, Rev 3, Aug 2023
2. OGN/OPS/SYST/012, Rev 0, April 2022

Submitted by:

सुमित कुमार वर्मा - वरिष्ठ प्रबंधक

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On:04/08/2023(16:50:46) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:2

no changes were envisaged.

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

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On:05/08/2023(09:27:41) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:3

Please route through all sectional incharges of EMD for comprehensive Review.

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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On:08/08/2023(11:16:40) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Checklist is reviewed. Following changes have been made:

SLNO	Existing description	Revised description
Checklist point 1	No person is standing in front of breaker at the time of planned switching	<i>No unauthorized person is standing in MCC room at the time of planned switching</i>
Checklist point 4	Automatic interlock switching checked at least once in last six months (Date of last checking -)	<i>Automatic interlock switching checked during every OH. (Date of last checking -)</i>

Revised LMI is attached.

Being routed to all sectional incharges.

Submitted by:

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On:28/08/2023(16:29:46) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:5

For routing through all sectional incharges of EMD for comprehensive Review in their area.

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On:31/08/2023(09:52:56) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:6

reviewed

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Note No:7

Please include the following CHP Switchgears also:

1. Wagon tippler MCC (under Annexure-A)
2. Wagon tippler Pump House MCC (under Annexure-D.)

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

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On:01/09/2023(09:41:54) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:8

LMI will be modified after routing to all sections.

Submitted by:

सुमित कुमार वर्मा - वरिष्ठ प्रबंधक

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On:01/09/2023(10:00:08) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:9

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On:04/09/2023(11:03:12) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:10

Submitted by:

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On:05/09/2023(10:04:17) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:11

Submitted by:

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On:06/09/2023(09:34:27) **Ref:**SMTTP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:12

Reviewed

Submitted by:

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On:11/09/2023(15:51:55) **Ref:**SMTTP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:13

Reviewed, inline with OGN 12, revised 06.04.2022

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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On:11/09/2023(17:07:32) **Ref:**SMTTP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:14

Review as discussed.

Submitted by:

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On:11/09/2023(17:21:29) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:15

CHP MCCs added as mentioned in note no 7.

Revised LMI is attached for approval.

Submitted by:

सुमित कुमार वर्मा - वरिष्ठ प्रबंधक

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On:11/09/2023(18:15:04) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:16

Reviewed and found in order

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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Note No:17

Submitted by:

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On:12/09/2023(09:05:59) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:18

Submitted by:

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On:12/09/2023(17:37:38) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:19

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:14/09/2023(10:00:11) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:20

Pl review the LMI (As per clause No 8 responsibility of implementation of this LMI rests with SCE/Operation also)

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

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On:14/09/2023(11:00:03) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:21

Proposed LMI is reviewed and found in line with OGN.

Submitted by:

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On:18/09/2023(14:14:42) **Ref:**SMTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:22

Forwarded for approval please

Submitted by:

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On:25/09/2023(21:07:02) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

Note No:23

Approved & Submitted by:

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On:26/09/2023(18:44:11) **Ref:**SMTTPP/ELECT MAINT/2023-24/LMIswitc/366136

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

NTPC LIMITED.
SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION
LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE: Aug 2023

**SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS
WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED**

APPROVED FOR

IMPLEMENTATION _ _ _ _ _

BY GROUP GENERAL MANAGER (SIMHADRI)

LMI/OGN/OPS / SYST / 012
 ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

NTPC LIMITED.
SIMHADRI SUPER THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED		
LMI No.	LMI/OGN/OPS / SYST / 012		
ISSUE NO.	2	Revision : 3	Date: Aug, 2023

	NAME	DESIGNATION
Prepared by	Sumit Kumar Verma	Sr. Manager (EM)
Checked by	A Bala Balaji	DGM (EM)
Checked by	T Vijay Krishna	AGM (EM)
Checked by	Dharmendra Kumar Tomar	HOS(EM)
Checked by	Mathew Eipe Kovoov	GM(Maintenance
Checked by	T.R. Rajagopalan	GM(Operation)
Reviewed by	Dhurjati Prasad Patra	GM(O&M)

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED

CONTENTS

S.NO.	DESCRIPTION	PAGE NO.
1.0	OBJECTIVE	4
2.0	SCOPE	4
3.0	REFERENCE DOCUMENT	4
4.0	SUPERCEDED DOCUMENT	4
5.0	THE SPECIFIC INSTRUCTIONS TO BE FOLLOWED	5
6.0	METHODOLOGY OF IMPLEMENTATION	5
7.0	FREQUENCY OF CHECKING AND MEASUREMENTS	5
8.0	THE RESPONSIBILITY CENTERS FOR IMPLEMENTATION	6
9.0	THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS	6
10.0	RECORD KEEPING	6
11.0	REVIEW	6
12.0	RECORDS OF REVISION	6
	ANNEXURE-A	7
	ANNEXURE-B	8
	ANNEXURE-C	10
	ANNEXURE-D	10
	ANNEXURE-E	12
	ANNEXURE-F	13
	CHECKLIST	15

SWITCHING OF POWER STATION DISTRIBUTION SYSTEMS WHERE FAULT LEVEL RATINGS MAY BE EXCEEDED

1.0 Objective :

During the normal startup of the unit, GT should be charged and 11KV auxiliary supply for unit to be taken from Unit transformer after changing over from standby transformer. Other boards may be paralleled for change over operations as per requirement as part of normal operations.

In such cases during the period for which the boards are paralleled, the fault level of the buses may exceed the designed fault level of the bus of switching breaker. For this period of time, extreme care has to be taken.

The aim of this LMI is to put forward the procedures and precautions to be followed during such switching operations so that damage to equipment and injury to personnel are avoided.

2.0 Scope :

This LMI is applicable to both the following switchboards of Simhadri St 1&2.:

- (i) All 33 kV switchboards
- (ii) All 11kV switchboards
- (iii) All 3.3kV switchboards
- (iv)** All 415V switchboards

3.0 Reference documents :

This is based on Operation guidance Note OGN/OPS/SYST/012.

4.0 Superseded Documents : LMI/OGN/OPS/SYST/012 ISSUE:2 REV:2 JUNE 2021

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

5.0 Specific instructions to be followed :

In order to minimize the risk to personnel during prolonged operation of switchgear at higher than normal fault levels, precautions should be taken. During the automatic/manual changeover, the paralleling is only momentary and is removed very fast. The risk during this operation is neglected. During changeover operations access to a switchboard or switch room which is subjected to excess fault level (as per annexure-E) shall be prohibited. The connection(s) between systems should be maintained for minimum possible time as is necessary, wherever such switching is carried out. Basically all planned changeovers are to be done by Operation dept. and cares as mentioned in previous section should be followed. To ensure the healthiness of auto changeover circuits, Electrical maintenance will take care of the scheduled checking.

6.0 Methodology of implementation :

The above procedure has to be adopted so that buses whose designed fault levels are exceeded temporarily can be operated safely. Basically all planned changeovers are to be done by Operation dept. and cares as mentioned in previous section should be followed. To ensure the healthiness of auto changeover circuits, Electrical maintenance will take care of the scheduled checking as per Annexure-F.

7.0 Frequency of checking:

Auto changeover and its interlock checking for all boards has to be carried out during unit overhauling/board PM.

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

8.0 Responsibility center for implementation:

1. Concerned section Heads of Electrical maintenance – Testing, Makeup water system, Switchyard, CHP groups.
2. SCE/Operation.

9.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOS(EM) and can be dispensed with his approval.

10.0 RECORD KEEPING:

Records will be kept with respective sections of Electrical Maintenance for healthiness checking of changeover circuits and for planned changeovers records will be maintained by Operation.

11.0 REVIEW :

Document will be reviewed after two years by HOP (Simhadri).

12.0 RECORDS OF REVISION:

1. Checklist
2. Annexure A and Annexure D
(changes are highlighted in blue color)

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

ANNEXURE-A

LIST OF SWITCHGEARS WITH AUTO-CHANGEOVER BUT WITHOUT SYNCHRONISATION.

Unit turbine switchgear-1KA
Boiler MCC-1HA
Station ACDB-1QA
Air washer MCC-1SA
AC & Ventilation MCC-1TA
ESP Switchgear-1DC
ESP Switchgear-1DD
ESP Switchgear-1DE
ESP Switchgear-1DF
Unit-1 ACP Pass-A
Unit-1 ACP Pass-B
Unit-1 ACP Pass-C
Unit-1 ACP Pass-D
Unit turbine switchgear-2KA
Boiler MCC-2HA
Station ACDB-2QA
Air washer MCC-2SA
AC & Ventilation MCC-2TA
ESP Switchgear-2DC
ESP Switchgear-2DD
ESP Switchgear-2DE
ESP Switchgear-2DF
Unit-2 ACP Pass-A
Unit-2 ACP Pass-B
Unit-2 ACP Pass-C
Unit-2 ACP Pass-D
Service building MCC-OTA
Offsite service switchgear-ODG
Fuel oil switchgear-ODC
PT Plant MCC
DM Plant MCC
CHP 3.3 KV MCC
CHP 415 V MCC-1
CHP 415 V MCC-2
CHP 415 V MCC-3,
CHP 415 V MCC-4

Wagon tippler MCC

ANNEXURE-B

**LIST OF FEEDERS WITH SYNCHRONIZATION CHANGEOVER THROUGH
 SYNCH CHECK FUNCTION IN ECD**

Unit service switchgear-1DA
 Station service switchgear-ODA
 Unit emergency switchgear-1DG
 11 KV Unit switchgear-1BA
 11 KV Unit switchgear-1BB
 3.3KV switchgear-1CA
 Unit service switchgear-2DA
 Station service switchgear-ODB
 Unit emergency switchgear-2DG
 11KV Unit switchgear-2BA
 11KV Unit switchgear-2BB
 3.3 KV switchgear-2CA
 3.3 KV Ash handling switchgear-OCA
 Unit Emergency switchgear#3
 Unit Emergency switchgear#4

**LIST OF FEEDERS WITH SYNCHRONIZATION CHANGEOVER THROUGH
 SYNCH CHECK FUNCTION IN-BUILT IN NUMERICAL RELAY IN SWITCHGEAR**

Unit service switchgear#3
 Station Service Switchgear#3
 Boiler MCC#3
 Turbine MCC#3
 Ash Extraction switchgear#3
 Air washer/Vent/AHU MCC#3
 Unit service switchgear#4
 Station Service Switchgear#4
 Boiler MCC#4
 Turbine MCC#4
 Ash Extraction switchgear#4

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

*Air washer/Vent/AHU MCC#4
ESP Switchgear#3A
ESP Switchgear#3B
ESP Switchgear#3C
ESP Switchgear#3D
ESP Switchgear#4A
ESP Switchgear#4B
ESP Switchgear#4C
ESP Switchgear#4D
DM/PT/Service water SWGR stage-II
Fuel oil Pump house Switchgear stage-II
Fire Water/CW Service switchgear Stage-II
Ash Water/TAC SWGR Stage-II
Ash Silo Service MCC Stage-II
3.3 KV CHP switchgear Stage-II
415V CHP MCC II-A
415V CHP MCC II-B
415V CHP MCC II-C
33KV MST switchgear
11 KV Sea water switchgear
415 V Sea water switchgear
415 V Sweet water switchgear
415 V Ash re-circulation MCC
11KV Unit Switchgear 3UA
11KV Unit Switchgear 3UB
11KV Station Switchgear SB
11KV Station Switchgear SC
3.3KV Switchgear 3UAA
11KV Unit Switchgear 4UA
11KV Unit Switchgear 4UB
11KV Station Switchgear SD
11KV Station Switchgear SE
3.3KV Switchgear 4UAA*

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

ANNEXURE-C

LIST OF LT FEEDERS IDENTIFIED FOR MANUAL CHANGEOVER WITH BREAKERS AS INCOMERS & BUS COUPLERS.

Ash compressor MCC-ODD
Ash slurry switchgear-ODF
Raw water MCC
Switchyard MCC
Stage-1 CW MCC
Admin Building SWGR
Sankalp Annex-2 SWGR
Samanvay Annex(Saavyas-2) Switchgear
ACP pass-3A,3B,3C,3D MCC
ACP pass-4A,4B,4C,4D MCC

ANNEXURE-D

LIST OF SWITCHGEARS IDENTIFIED FOR MANUAL CHANGEOVER WITH SFUs AS INCOMERS & BUSCOUPLERS.

Turbine ACDB-1KC
Boiler ACDB-1HD
Boiler Damper MCC-1HC
Coal Mill MCC-1HC
Coal Mill MCC-1HE
ESP MCC-1HF
ESP AC & Ventilation MCC-1TB
Ash Silo MCC-OSI
Unit 1 Soot Blower MCC
Turbine ACDB-2KC
Boiler ACDB-2HD
Boiler Damper MCC-2HC
Coal Mill MCC-2HE

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

ESP MCC-2HF
ESP AC & Ventilation MCC-2TB
Stage 1 Chlorination MCC OCL
Stage 1 CW Pump house MCC OCW
Stage 1 Clarified water pump house MCC OSB
Stage 1 Fire Water Pump House MCC OSC
Stage 1 Aux Boiler MCC OSA
Unit 2 Soot Blower MCC
Unit 3 Station ACDB
Unit 3 Boiler ACDB
Unit 3 Boiler Valve DB
Unit 3 Turbine Valve DB
Unit 3 Bottom Ash MCC
Unit 3 Soot Blower MCC
Unit 3 ESP/ID Fan Emergency MCC
Unit 3 ESP/VFD AC MCC
Unit 4 Station ACDB
Unit 4 Boiler ACDB
Unit 4 Boiler Valve DB
Unit 4 Turbine Valve DB
Unit 4 Bottom Ash MCC
Unit 4 Soot Blower MCC
Unit 4 ESP/ID Fan Emergency MCC
Unit 4 ESP/VFD AC MCC
Switchyard ventilation MCC
Switchyard AC MCC
CW ventilation MCC
ACDBs in Sea Water
ACDBs in Sweet water
Ash re-circulation system ACDB
Effluent disposal system MCC
Effluent treatment plant MCC-1
Effluent treatment plant MCC-2
Effluent treatment plant MCC-3
Effluent treatment plant MCC-4
Effluent treatment plant MCC-5
Effluent treatment plant MCC-6
Hydrogen Plant MCC OSD

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

Workshop MCC
Stage 2 CPU MCC
Stage 2 Effluent Treatment MCC
Stage 2 CW MCC
Stage 2 AC MCC
Stage 2 Emergency AC MCC
Wagon tripler pump house MCC

ANNEXURE-E

FAULT LEVEL OF BUSES DURING PARALLELING OF BUSES FOR CHANGE OVER

Basic design fault level data for buses as per technical specification :

SL.NO.	BUS	DESIGN FAULT LEVEL	FAULT WITHSTAND TIME
1	400 kV	40kA	1 Sec
2	33 kV	12.5 kA	1 Sec
3	11 kV	40 kA	1 Sec
4	3.3 V	40 kA	1 Sec
5	415 V	45 kA	1 Sec

Actual fault level during paralleling/changeover :

SL. NO.	BUS	FAULT LEVEL DURING PARALLELING	REMARKS
1	33kV switchgear	4.403 kA	
2	1BA/1BB/2BA/2BB	43kA	Fault level exceeds designed fault level.
3	1CA/2CA	36.5kA	
4	OCA	30.6kA	
5	415V fed from 2MVA transformers	55.9kA	Fault level exceeds designed fault level.
6	415V fed from 1.6MVA transformers	55.9 kA	Fault level exceeds designed fault level.
7	415V fed from 1MVA transformers.	47.15kA	Fault level exceeds designed fault level.

LMI/OGN/OPS / SYST / 012
ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

ANNEXURE-F

METHODOLOGY OF IMPLEMENTATION :

SL. NO.	JOB DESCRIPTION	RESPONSIBILITY	ACTION TO BE TAKEN	FREQUENCY	LIST OF SWITCHGEARS
1	Switchgears having auto changeover schemes in LT system but without synchronization	Electrical Maintenance	Checking for blocking of auto Changeover scheme healthiness with master trip relay simulation.	In Unit Overhauling	As per Annexure-A
		Operation	During any planned changeover no additional drives should be started in changeover period. Before any planned changeover the healthiness of trip circuit supervision relay should be checked for the incomer/bus-coupler to be tripped.	To be taken care in all planned changeover.	
2	LT & HT switchgears having synchronization changeover scheme	Electrical Maintenance	Checking for blocking of changeover scheme healthiness with master trip relay simulation.	In Unit overhauling	As per Annexure-B
		Operation	During any planned changeover no additional drives should be started in changeover period. Before any planned changeover the healthiness of trip circuit supervision relay should be checked for the incomer/bus coupler to be tripped.	To be taken care in all planned changeover.	

LMI/OGN/OPS / SYST / 012
 ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

3	LT switchgears with manual changeovers having breakers.	Operation	During any planned changeover no additional drives should be started in changeover period. Before any planned changeover the healthiness of trip circuit supervision relay should be checked for the incomer/bus coupler to be tripped.	In all planned changeover	As per Annexure-C
4	LT Switchgears with SFU as incomer & buscoupler.	Operation	During any planned changeover no additional drives should be started in changeover period. Before any planned changeover the healthiness of feeder to be switched on. The person should ensure that the feeder required properly switched off.	In all planned changeover	As per Annexure-D
5	HT Switchgears manual changeover		Not allowed		

LMI/OGN/OPS / SYST / 012
 ISSUE NO.: 2, REVISION-3, DATE : Aug, 2023

CHECKLIST

Sl No	Checks	Status(Yes/no)
1	<i>No unauthorized person is standing in MCC room at the time of planned switching</i>	
2	<i>The two supply sources being coupled are healthy</i>	
3	<i>In any case the coupling is not for more than 1.2 seconds.</i>	
4	<i>Automatic interlock switching checked during every OH. (Date of last checking -)</i>	
5	<i>No switching operation except coupling and decoupling is being carried out</i>	